

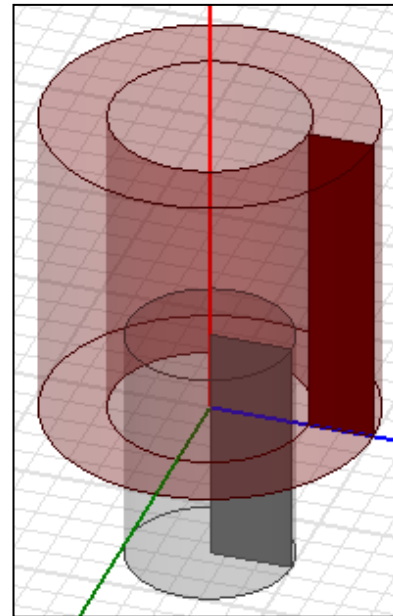
Workshop 1.1: Magnetostatic 2D Analysis

Release 2020R2

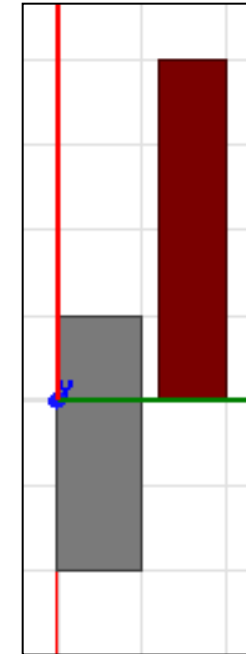


Overview

- Background
 - This workshop assumes little or no prior experience of Maxwell. Basic principles will be demonstrated through the creation of a simple cylindrical symmetry.
- Force calculation in Magnetostatic Solver
 - This workshop will discuss how to set up a force calculation in the 2D Magnetostatic Solver.
- Problem Description
 - As shown in the following picture, a coil and slug are drawn in a plane using RZ symmetry. The coil carries a current exerting a vertical force on the ferro-magnetic slug.



Actual 3D Coil and Slug



2D Symmetric Coil and Slug
about z-axis

Model Setup

- Insert Design

- Select the menu item **Project** → **Insert Maxwell 2D Design**, or click on the icon in drop down list Maxwell on panel Desktop

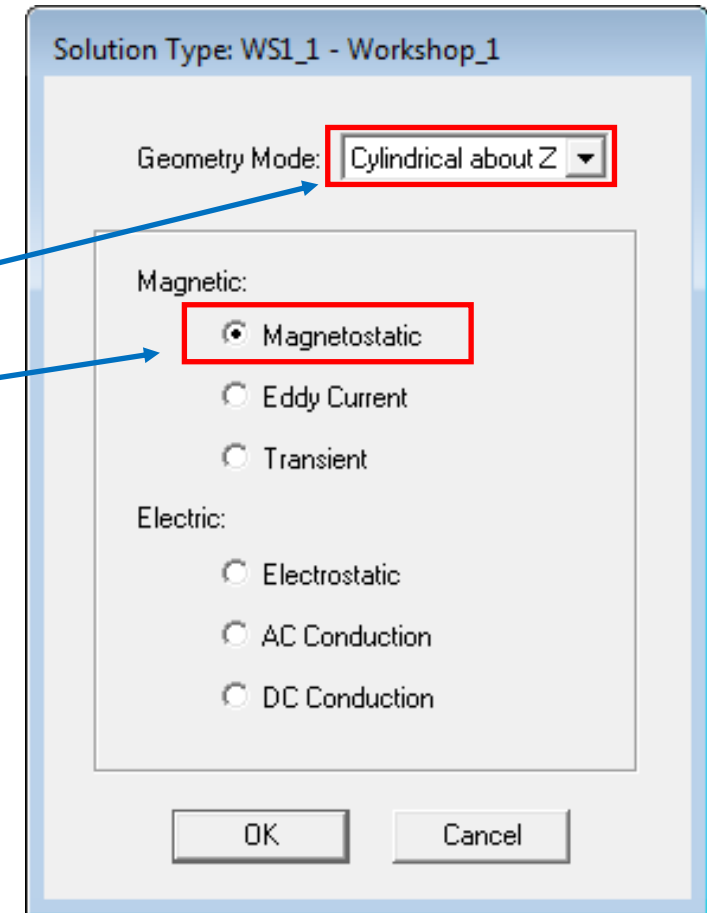


- Set Solution Type

- Select the menu item **Maxwell 2D** → **Solution Type**
- Geometry Mode: **Cylindrical about Z**
- Choose **Magnetic** → **Magnetostatic**
- Click the OK button

- Set Model Units

- Select the menu item **Modeler** → **Units**
 - Select units: **mm** (millimeters) and press OK



Create Model

- Create Slug
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: -10, Press the **Enter** key
 - Using the coordinate entry fields, enter the opposite corner
dX: 5, dY: 0, dZ: 15, Press the **Enter** key
 - Change the name of resulting sheet to **Slug** and color to **Gray** using the property window
 - Change the material of the sheet to **Steel_1008** by selecting the object **Slug** and going to menu item **Modeler** → **Assign Material**

X: 0 Y: 0 Z: -10 Absolut Cartesian mm

dX: 5 dY: 0 dZ: 15 Relative Cartesian mm

Name	Value	Unit	Evaluated Value
Name	Slug		
Material	"steel_1008"		"steel_1008"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Display Wi...	☐		
Color			
Transparent	0		

Attribute

Create Model

- Create Coil
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
 - **X: 6, Y: 0, Z: 0**, Press the **Enter** key
 - Using the coordinate entry fields, enter the opposite corner
 - **dX: 4, dY: 0, dZ: 20**, Press the **Enter** key
 - Change the name of resulting sheet to **Coil** and color to **Brown** using the property window
 - Change the material of the sheet to **Copper** by selecting the object **Coil** and going to menu item **Modeler** → **Assign Material**

X: Y: Z: Absolut Cartesian mm

dX: dY: dZ: Relative Cartesian mm

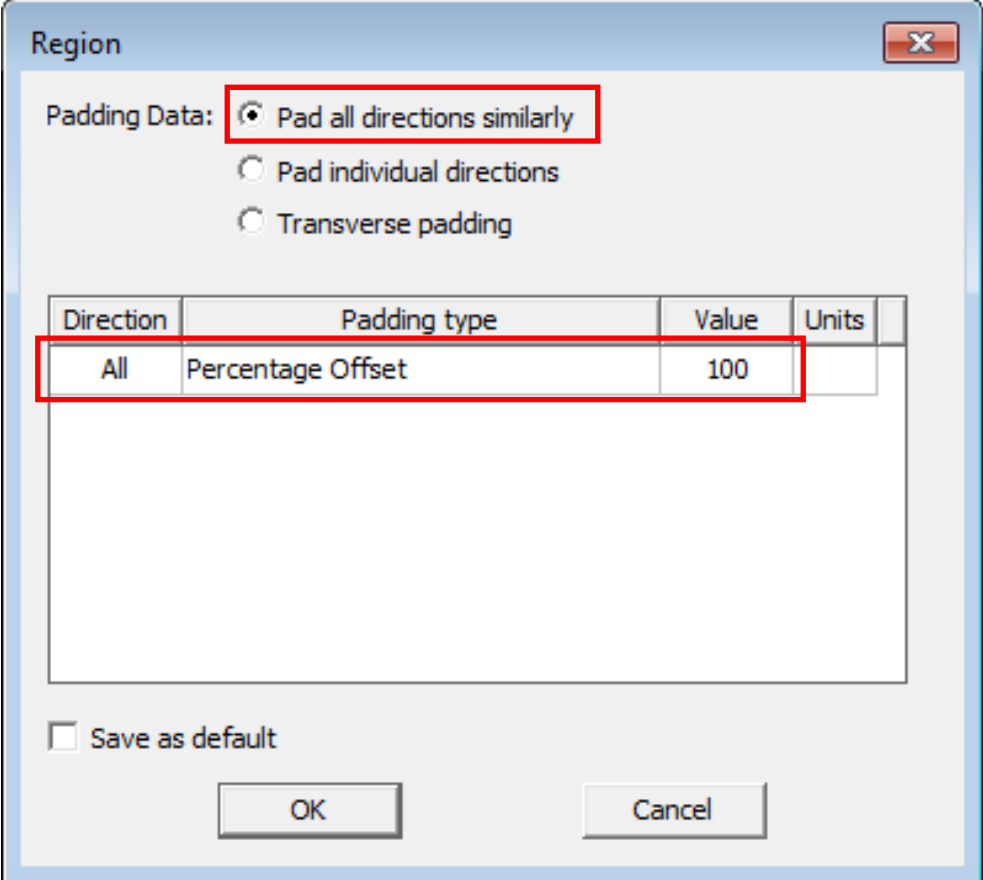
Name	Value	Unit	Evaluated Value
Name	Coil		
Material	"copper"		"copper"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Display Wi...	☐		
Color	Brown		
Transparent	0		

Attribute

Define Region

- Create Simulation Region
 - Select the menu item **Draw** → **Region** or click on the icon 
 - In Region window
 - Pad all directions similarly: ☒ **Checked**
 - Padding Type: Percentage Offset
 - Value: 100
 - Press OK

Note: the region extension in negative X direction is neglected due to RZ-symmetry about the Z-axis.



The image shows a 'Region' dialog box with the following settings:

- Padding Data:** ☒ Pad all directions similarly (highlighted with a red box)
- ☐ Pad individual directions
- ☐ Transverse padding

Direction	Padding type	Value	Units
All	Percentage Offset	100	

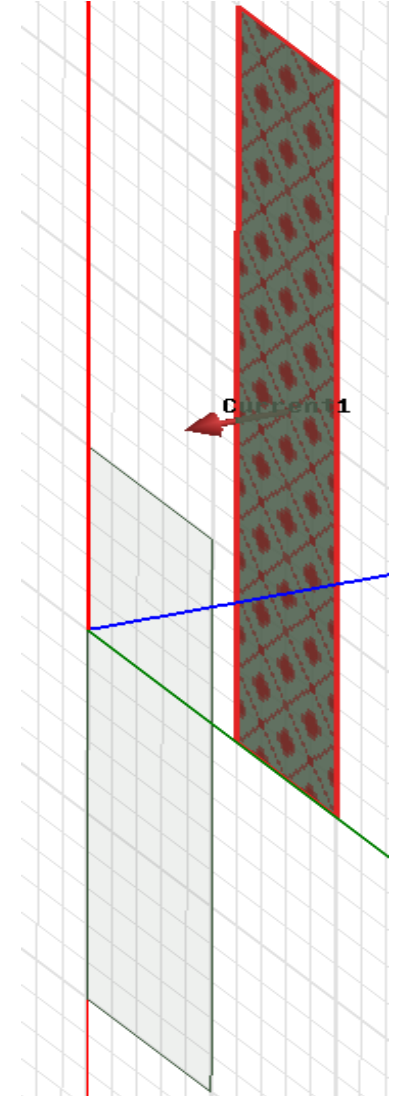
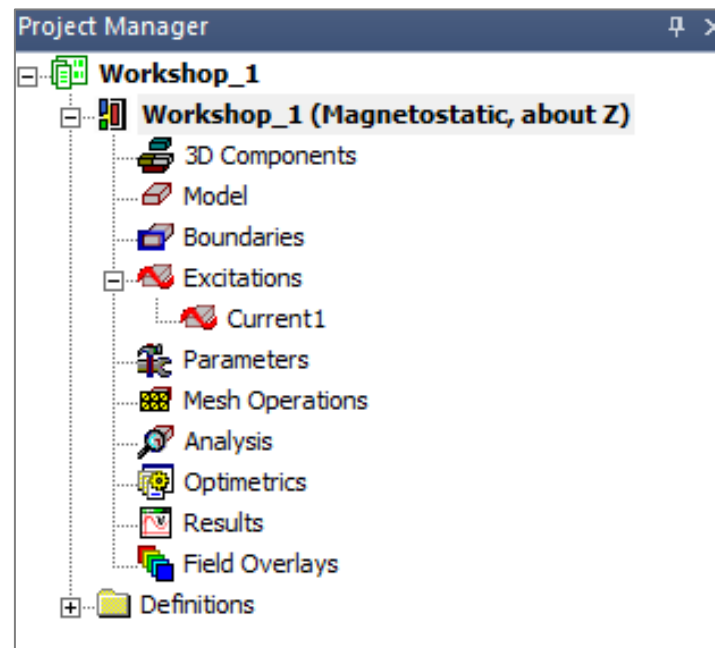
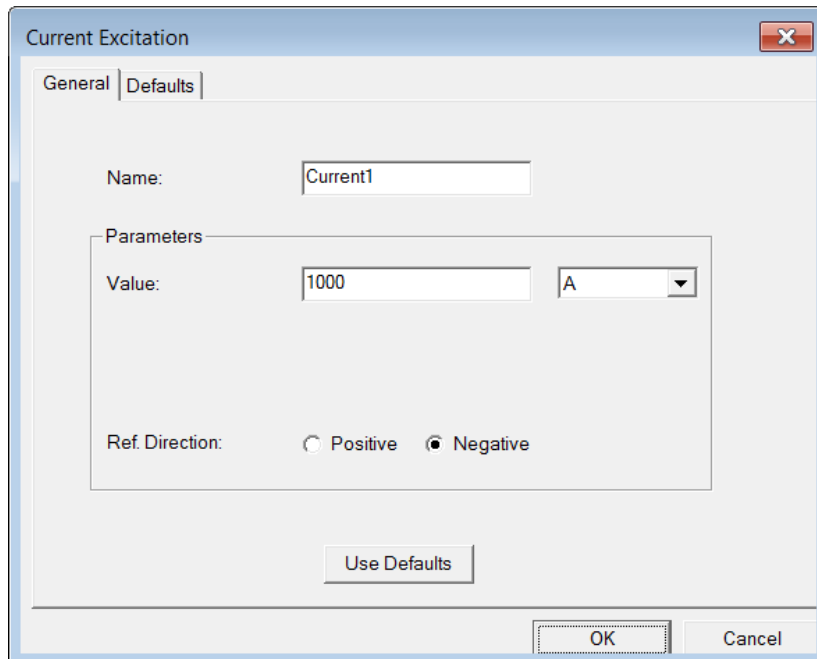
The first row of the table is highlighted with a red box.

☐ Save as default

OK Cancel

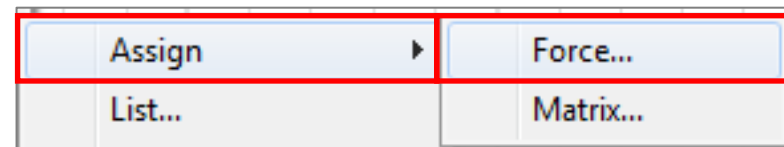
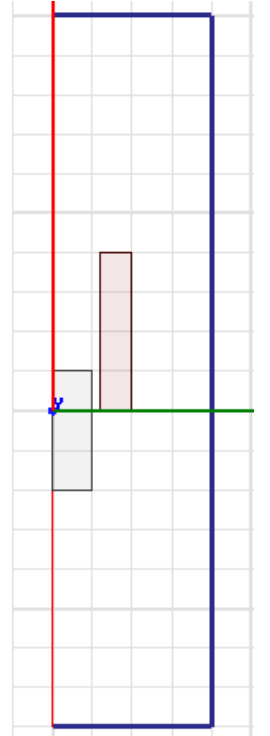
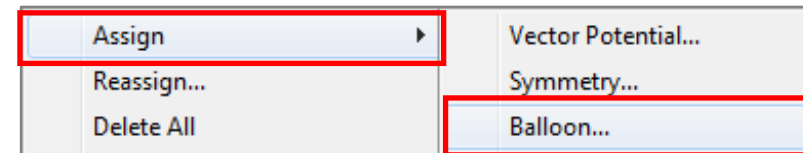
Assign Excitations

- Assign Excitations
 - Select the sheet **Coil** from history tree
 - Select the menu item **Maxwell 2D** → **Excitations** → **Assign** → **Current**
 - Name: **Current1**
 - Value: **1000**
 - Ref. Direction: **Negative** (current will be in the negative Y direction)
 - In the Project Manager window under Excitations a new item called “Current1” is present



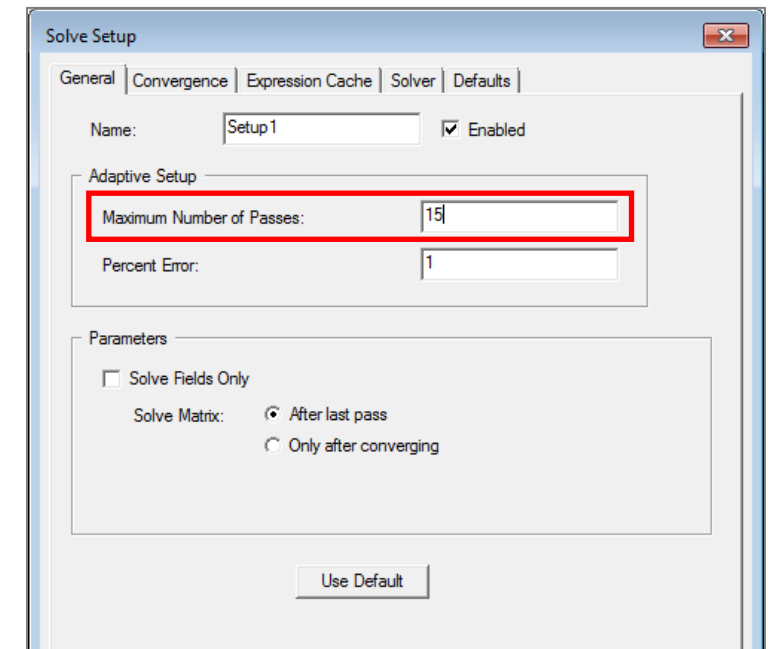
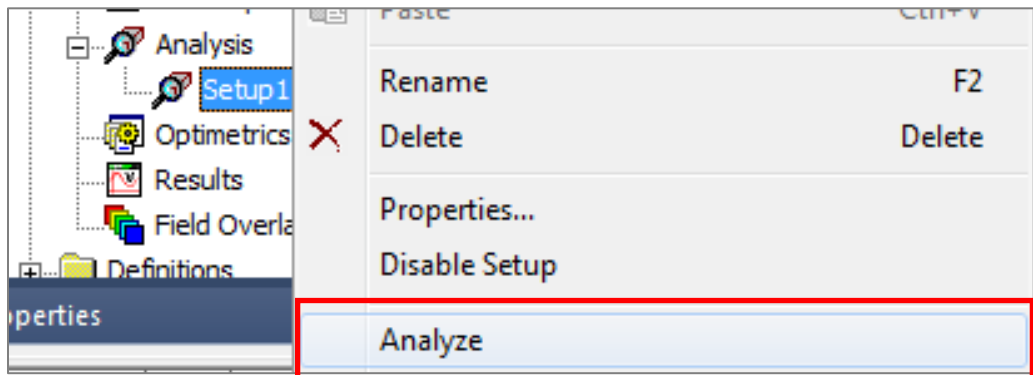
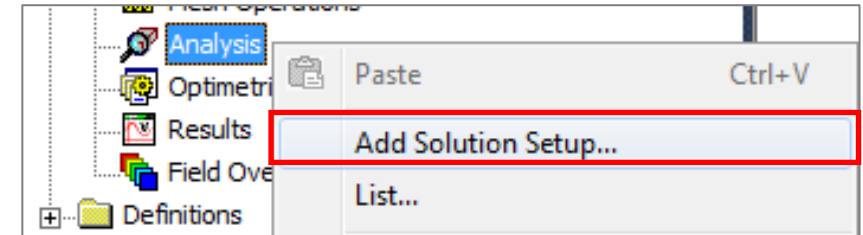
Assign Boundary Conditions and Force Parameter

- Assign Boundary Conditions to Region Edges
 - Select the object **Region** from history tree
 - Select the menu item **Edit → Select → Edges**
 - Select the three edges of the Region
 - Select the menu item **Maxwell 2D → Boundaries → Assign → Balloon**
 - In Balloon Boundary window, press OK
- Assign Force Calculation
 - Select the object **Slug** from history tree
 - Select the menu item **Maxwell 2D → Parameters → Assign → Force**
 - In Force Setup window, press OK



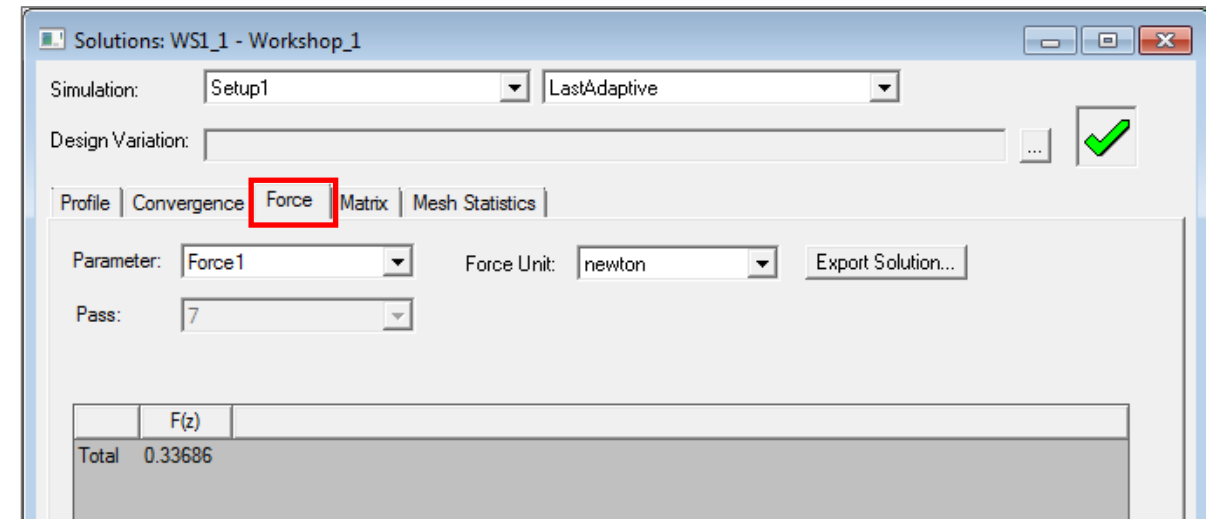
Analyze

- Create an analysis setup:
 - Select the menu item **Maxwell 2D** → **Analysis Setup** → **Add Solution Setup**
 - **Solution Setup Window:**
 - **General Tab**
 - **Maximum Number of Passes: 15**
 - **Click the OK button**
- Start the solution process:
 - In the Project Manager window **RMB on Setup1** → **Analyze**



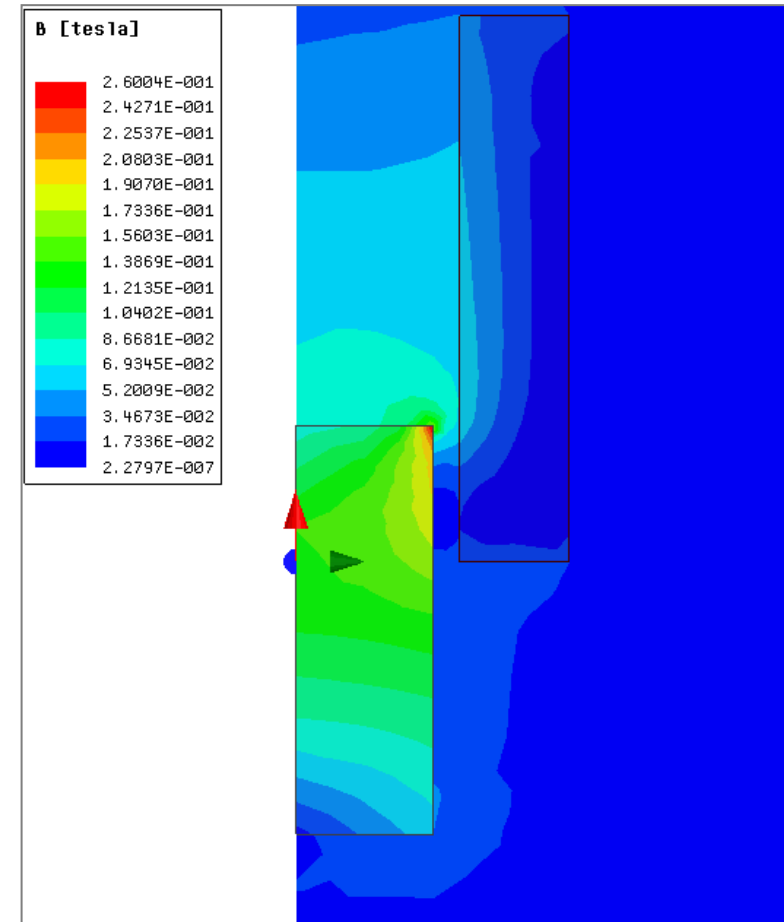
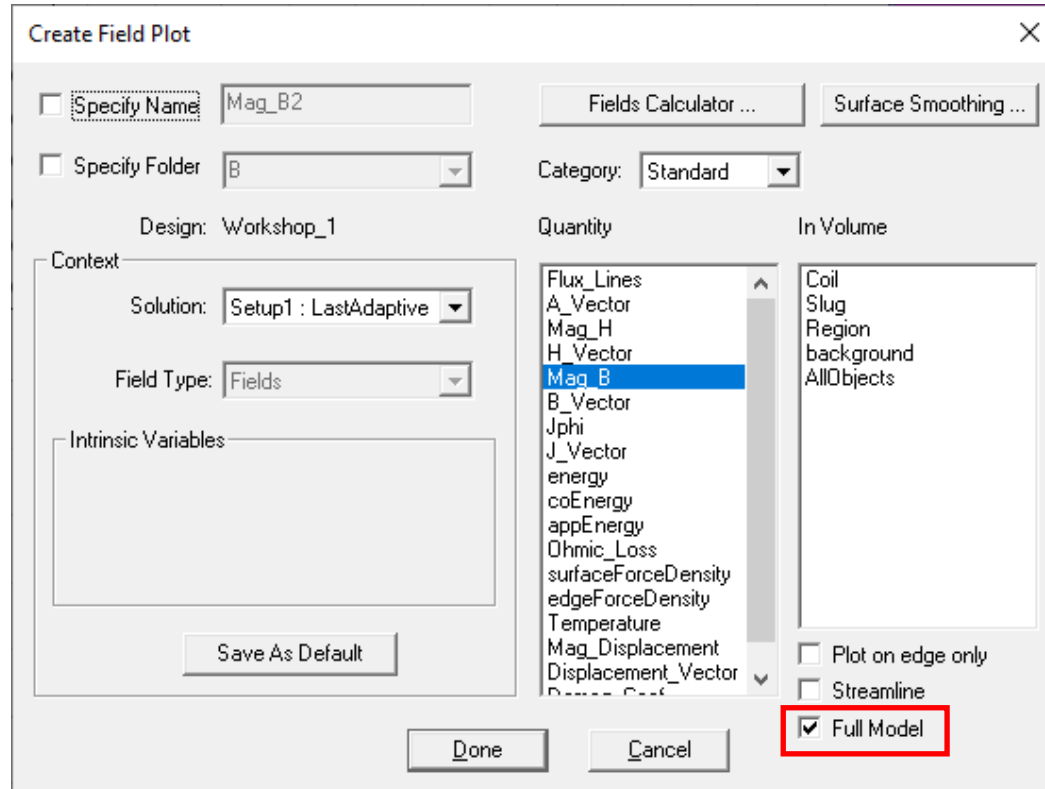
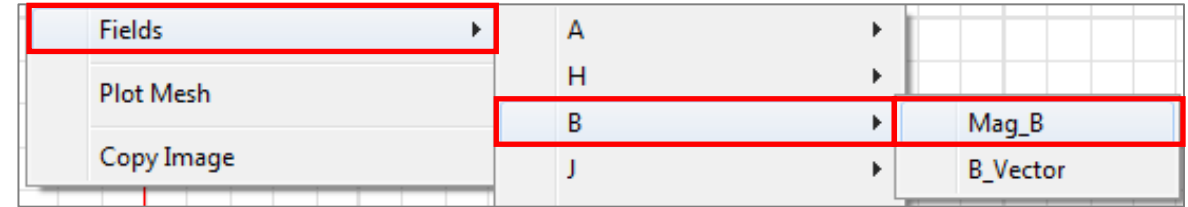
Note: before starting the simulation, Maxwell asks the user to save the Project, specifying name and path

- To View Calculated Force Value
 - Select the **Force** tab



Create Field Plot

- Plot Magnetic Flux Density
 - Press **Ctrl+A** to select all Objects
 - Select the menu **Maxwell 2D** → **Fields** → **Fields** → **B** → **Mag_B**
 - In Create Field Plot window, uncheck **Full Model**. Press Done



Saving the Project

- This completes the workshop
- Save the file with the name **Workshop_1_1** in the working folder



End of Presentation